



Student Name: _____

Teacher Code: HEWP THOS LOPF TRIA AJIA CHEY

Test Name: KINEMATICS - PARTY QUIZ SHOWDOWN

MATHEMATICS TEST

UNIT 1/2

Test theme: KINEMATICS

Reading time: 5 minutes

Writing time: 50 minutes

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
5	5	25

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your name in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer all questions in the spaces provided. Calculus restriction: use only the power rule for differentiation and integration of powers of x . Unless otherwise specified, an exact answer is required. Appropriate working must be shown for questions worth more than 1 mark.

Round 1: The Number Ones

The strobe lights hit the stage as Felix adjusts his sparkly blazer and glances at the scoreboard. "Welcome back to the Party Quiz! It is a tight race, though The Number Ones are trailing so far behind I am not sure they are even in the same building anymore. Selby, buddy, you have got to pick up the pace!" Felix points to a motorised disco ball sliding along a horizontal rail. "Number Ones, if you miss this, you are staying in dead last!"

Question 1 (4 marks)

The position $x(t)$ of the disco ball, in metres, relative to the centre of the stage is

$$x(t) = t^3 - 9t^2 + 24t - 16, \quad t \geq 0$$

a. Derive expressions for the velocity $v(t)$ and acceleration $a(t)$.

1 mark

b. Find the times t when the disco ball is instantaneously at rest.

2 marks

c. Determine the acceleration of the ball at the moment it passes the centre of the stage ($x = 0$) for the first time.

1 mark

Round 2: Team Two

"Next up, Team Two!" Felix shouts, firing a confetti cannon over their heads. "The recoil of this cannon is legendary. If you want those bingo points, tell me how far the trigger mechanism slides back!" A teammate trips over a cable and falls backward with a massive splash into the adjacent pool. Felix does not stop the clock. "Ignore the drowning teammate! Keep calculating!"

Question 2 (5 marks)

The acceleration of the confetti cannon's trigger is

$$a(t) = 18t - 24, \quad t \geq 0$$

At $t = 1$, the trigger is moving with velocity -2 m/s at a position of 15 m.

a. Derive the expression for the velocity $v(t)$ of the trigger.

2 marks

b. Calculate the total distance travelled by the trigger between $t = 0$ and $t = 2$.

3 marks

b. Prove that Joker Kermond changes direction at $t = 6$. Hence, find the total distance he has travelled vertically during the first 8 seconds of his dive.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

The Big Red Buzzer: Team MBEK

"Team MBEK, you are in the lead, do not choke now!" Felix bellows. "It is a race to the Big Red Buzzer. One of you is accelerating, the other has a head start. I want to know exactly what the motion tells us." The members of MBEK line up with ruthless efficiency, eyes fixed on the prize.

Question 4 (5 marks)

Two contestants from MBEK, Player M and Player K, run along a straight track. Their positions are

$$x_M(t) = 2t^2 - 4t + 12, \quad x_K(t) = t^3 + 4.$$

- a. Determine whether there is any time $t > 0$ at which the two players are moving at the same velocity. Justify your answer.

2 marks

- b. Determine the distance between the two players at the moment Player K's acceleration is exactly triple Player M's acceleration.

3 marks

Final Round: The Trophy

Felix stands in the centre of the stage as the final scores are tallied. "One last question for the crown. The prize trophy is being lowered from the ceiling. Caonima, MBEK, Team Two, Number Ones - lock it in! MBEK is looking for the sweep, while the Number Ones are just looking for their dignity!"

Question 5 (4 marks)

A trophy is lowered such that its velocity is

$$v(t) = k(t - 5)^2, \quad 0 \leq t \leq 5.$$

At $t = 0$, the trophy is at a height of 50 m ($x = 50$).

a. Show that

$$k = -\frac{6}{5}$$

if the trophy reaches the ground ($x = 0$) at $t = 5$.

2 marks

b. Find the acceleration of the trophy at the instant it is exactly 25 m above the ground.

2 marks

"LOCK IN YOUR ANSWERS!"
CONGRATULATIONS TO MBEK FOR THE VICTORY.
COMMISERATIONS TO THE NUMBER ONES FOR... WELL, EVERYTHING.
WE HOPE TEAM TWO HAS DRIED OFF.

END OF TEST

ASSESSMENT GUIDE / MARKING SCHEME

Specialist Mathematics Units 1/2 - Kinematics: Felix's Party Quiz Showdown

Award marks for mathematically equivalent working. Consequential working may be credited where appropriate. Exact values are required unless otherwise indicated.

Question 1 - 4 marks

- | | |
|---|----------|
| a. Differentiates correctly: $v(t) = 3t^2 - 18t + 24$ and $a(t) = 6t - 18$. | 1 |
| b. Sets $v(t) = 0$ and factorises $3(t-2)(t-4)=0$. | 1 |
| Obtains $t = 2$ s and $t = 4$ s. | 1 |
| c. Factorises $x(t)=(t-1)(t-4)^2$, so the first centre crossing is $t=1$; hence $a(1)=-12$ m/s ² . | 1 |

Question 2 - 5 marks

- | | |
|--|----------|
| a. Integrates acceleration: $v(t) = 9t^2 - 24t + C$. | 1 |
| Uses $v(1)=-2$ to obtain $C=13$, hence $v(t)=9t^2-24t+13$. | 1 |
| b. Finds the velocity zeros in $[0,2]$: $t = (4 \pm \sqrt{3})/3$. | 1 |
| Uses an antiderivative of velocity, $x(t) = 3t^3-12t^2+13t+C$, and splits the interval at both turning times. | 1 |
| Adds the absolute displacements over the three intervals to obtain total distance $2 + 8\sqrt{3}/3$ m. | 1 |

Question 3 - 7 marks

a. Finds displacement from 0 to 4: $\int_0^4 \frac{1}{2}t^2 dt = 32/3$.	1
Finds displacement from 4 to 7: $\int_4^7 (24-4t) dt = 6$.	1
Combines the displacements correctly: total displacement = 50/3.	1
Hence obtains $x(7) = 100 + 50/3 = 350/3$ m under the stated signed-position convention.	1
b. Shows $v(6)=0$ and that v changes sign from positive for $4 < t < 6$ to negative for $6 < t \leq 8$.	1
Finds distances $0-4 = 32/3$, $4-6 = 8$, and $6-8 = 8$.	1
Total distance = $32/3 + 8 + 8 = 80/3$ m.	1

Question 4 - 5 marks

a. Equates velocities: $4t-4 = 3t^2$, giving $3t^2-4t+4=0$.	1
Notes the discriminant is $16-48=-32 < 0$, so there is no real time $t > 0$ at which the velocities are equal.	1
b. Finds $a_M=4$ and $a_K=6t$; solves $6t=3(4)$ to obtain $t=2$.	1
Evaluates positions: $x_M(2)=12$ and $x_K(2)=12$.	1
Distance between the players = 0 m.	1

Question 5 - 4 marks

a. Integrates velocity to express the displacement from 0 to 5: $\int_0^5 k(t-5)^2 dt = 125k/3$. **1**

Uses displacement -50 m to obtain $125k/3 = -50$, hence $k = -6/5$. **1**

b. Obtains height $x(t) = -2/5(t-5)^3$, and solves $x=25$ to get $t-5 = -5/\sqrt[3]{2}$. **1**

Differentiates velocity: $a(t) = 2k(t-5)$, giving $a = 12/\sqrt[3]{2} = 6\sqrt[3]{4} \text{ m/s}^2$. **1**

TOTAL: 25 MARKS